Notebook Summary

1. Dataset Overview

The dataset used in this analysis contains various details about movies, including:

- **Movie Details**: Title, release date, and runtime.
- **Financial Metrics**: Budget and revenue figures.
- **Popularity Metrics**: Popularity scores and vote counts.
- **Casting Information**: Names of the main cast members.
- **Production Details**: Production companies and directors.
- **Genre Classification**: Movie genres.
- **Additional Information**: Taglines, keywords, and a brief overview of the movie.

2. Data Cleaning

- Dropped unnecessary columns: `id`, `imdb_id`, `cast`, `homepage`, `tagline`, `keywords`, `overview`, and `runtime`.
- Duplicates were eliminated to guarantee data integrity.
- handled missing values and removed entries with a budget of zero.
- expanded the production_companies and genres columns to handle films with several production companies or genres.
- To fix the anomalies, the median budget was substituted for zero values in the budget.

3. Data Wrangling

- Calculated profit for each movie.
- Converted the `release_date` to a datetime format and extracted the release year.

4. Exploratory Data Analysis (EDA)

- **Scatter Plot of Budget Over the Years**: Visualised the trend of movie budgets over the years to make sure that the zero-budget entries were removed.
- **Profit vs. Budget Scatter Plot**: Visualised the trend of movie budgets over the years to make sure that the zero-budget entries were removed.

5. Group-by Plots

- Created a function `groupby_plots` to generate bar plots for various group-by analyses.
- Created a function `groupby_scatterplot` to generate scatter plots for various group-by analyses.
- **Most Popular Genre**: Identified the genre with the highest total popularity.
- **Average Profit for Each Genre**: Calculated and visualised the average profit per genre.

- **Top 10 Production Companies by Profit**: Identified and visualised the top 10 production companies based on total adjusted profit.

6. Correlation Analysis

- Created a function `plot_and_correlate` to calculate the Pearson correlation coefficient and generate a scatter plot with a regression line.
- Analysed the correlation between budget and profit.

Conclusion

1. **Most Popular Genre**:
    - The genre with the highest total popularity was identified.
2. **Average Profit for Each Genre**:
    - The average profit per genre was calculated, showing which genres tend to be more profitable on average.
3. **Top 10 Production Companies by Profit**:
    - The top 10 production companies based on total adjusted profit were identified.
4. **Correlation Between Budget and Profit**:
    - The correlation between budget and profit was calculated using the Pearson correlation coefficient, providing insight into the relationship between the amount spent on making a movie and the profit it generates.

These analyses offer a thorough grasp of the different facets of the film collection, assisting in the discovery of patterns and connections within the information.